SAILAAB · Punjab Flood Intelligence

Open SAR flood mapping · decade hazard atlas · impact analytics · district flood forecasting · live monitoring

ਸੈਲਾਬ · ਪੰਜਾਬ ਦੀ ਹੜ੍ਹ ਖੁਫ਼ੀਆ ਪ੍ਰਣਾਲੀ

LIVE: bakathefish.github.io/Flood   CODE + DATA: github.com/bakathefish/Flood

11	**467**	**105,183**	**91**	**ρ=0.72**	**0**	**0**
MONSOONS MAPPED	FLOOD DAYS PROCESSED	HA FLOODED 2025 (TIER-A)	TEHSILS SCORED	VS GOVT GIRDAWARI	CWC STATIONS IN PUNJAB	LOGINS REQUIRED

01 The problem

In August and September 2025, Punjab suffered its worst flood since 1988: all 23 districts affected, ~3.55 lakh people, 1.48–1.75 lakh hectares of crops destroyed. Four failures made it worse than it had to be. Nobody was forecasting: in the Central Water Commission's own state-wise table, Punjab has zero flood-forecasting stations, absent from a 226-station national network (Haryana 1, J&K 3; the one site CWC ever added on the Ravi is defunct). Punjab's rivers sit outside India's flood-forecast network. Maps existed but were locked: ISRO's NDEM produced excellent flood maps as static PDFs, one snapshot at a time, not analyzable, not open. Nobody knew the recurrence: Punjab has no public flood-frequency atlas (NRSC published them for Assam and Bihar), so villages that flooded four times in a decade were treated as first-time surprises. Relief ran on foot: girdawari compensation required weeks of manual crop surveys while satellite data that could target it sat unused.

02 What Sailaab is

A five-module open pipeline that turns free satellite radar into flood intelligence a district officer can act on. I built it in Punjab during the 2026 monsoon, and it is running live today (the monitor executes on public CI every 6 hours with zero accounts or keys; jurors can watch it commit).

MODULE	WHAT IT PRODUCES
2025 Flood Atlas	Sentinel-1 SAR flood extent (physics baseline + Random Forest), per-district and per-tehsil statistics
Decade Hazard Atlas	Every monsoon 2015–2025 mapped identically → Punjab's first public flood-frequency map + named "repeat victims" tehsil list
Impact Engine	Crop damage (₹351–523 crore band), population exposure, submergence-duration atlas, the rain×dams×releases causal chart
Forecaster	XGBoost district flood-risk model trained on the pipeline's own decade of SAR-derived labels, SHAP-explained and conformal-bounded
Live Monitor	Every new Sentinel-1 pass → district flood km² → ਪੰਜਾਬੀ / हिन्दी / English alerts, automatically

91 tehsils are individually scored, giving relief a named list: Khadur Sahib (Tarn Taran) flooded in 6 of the last 11 monsoons, Sultanpur Lodhi (Kapurthala) in 5, Moonak (Sangrur) and Patti (Tarn Taran) in 4. These are the exact administrative units where girdawari verification and pre-positioning should start. Population exposure adds the human denominator: 0.76–1.78 lakh people lived inside the mapped water (GHSL 2025), consistent with the official 3.55 lakh "affected" once you see that flooded land is overwhelmingly low-density cropland (146–206 people/km² vs Punjab's ~550 mean). Everything is login-free and reproducible: anonymous Planetary Computer STAC for Sentinel-1, keyless Copernicus-GFM WMS, IMD rainfall, CWC reservoir records. A district collector, a journalist, or a student can re-run every number with zero accounts.

03 The AI, and why it is honest

Random Forest flood classifier. Features: VV/VH backscatter, pre/post change, terrain. Labels: bootstrapped from *agreement* between our physics-threshold map and Copernicus GFM. No hand digitization, and we say so. Validation is spatial cross-validation across river basins (train Ravi–Beas districts → test Sutlej districts, and swap). We report the trap honestly: agreement-strata points are near-separable (fold F1 ≈ 1.0), so we publish the independent fresh-point comparison (OA 0.983 vs GFM) and the three-method envelope (Tier-A 34k < RF 52k < GFM 86k ha in-district). A small U-Net trained on the Sen1Floods11 hand-labeled benchmark (test IoU 0.63) completes a three-method benchmark table in which the target-domain RF wins and the imported deep model's resolution gap is quantified rather than hidden.

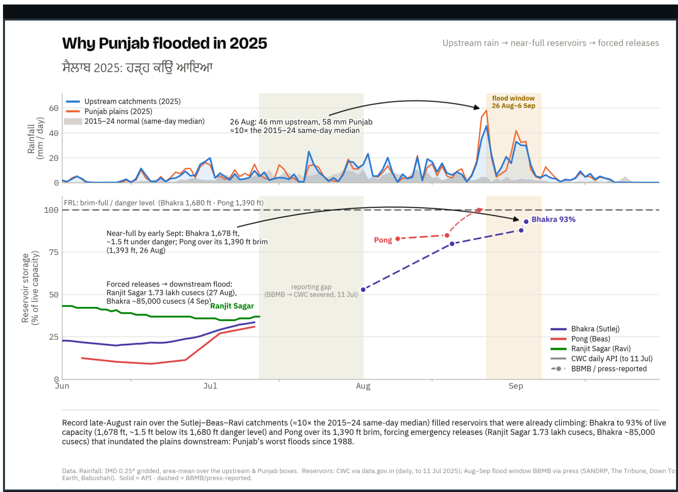
The self-labeling loop. Nobody hands you 11 years of Punjab flood labels, so the mapping engine manufactured its own: 1,177 day-slots probed, 467 flood-active days rasterized, 2,420 district-window training rows. Doing that exposed a trap that would silently poison any naive attempt: June and July "floods" in Punjab are largely rice transplanting, because deliberately flooded paddies inflate apparent flood area ~20×. All forecaster training excludes the transplant windows; calibrated late-season products ship alongside raw ones.

XGBoost district forecaster. Unit: district × 10-day window. Predictors: IMD rainfall (local + Himalayan upstream, lagged), reservoir storage and its rate of change, antecedent flooding, decade-frequency prior. Gradient boosting rather than deep learning: the sample-efficient, interpretable right tool for 2.5k tabular rows with 27 events. Leave-one-year-out ROC-AUC 0.946 (PR-AUC 15× base rate), leakage-checked with fold-safe priors; split-conformal intervals published, including their honest failure on the +10σ 2025 event (non-exchangeability, quantified). A pre-registered ERA5 sub-daily-intensity challenger *lost* to this model and the negative result is documented.

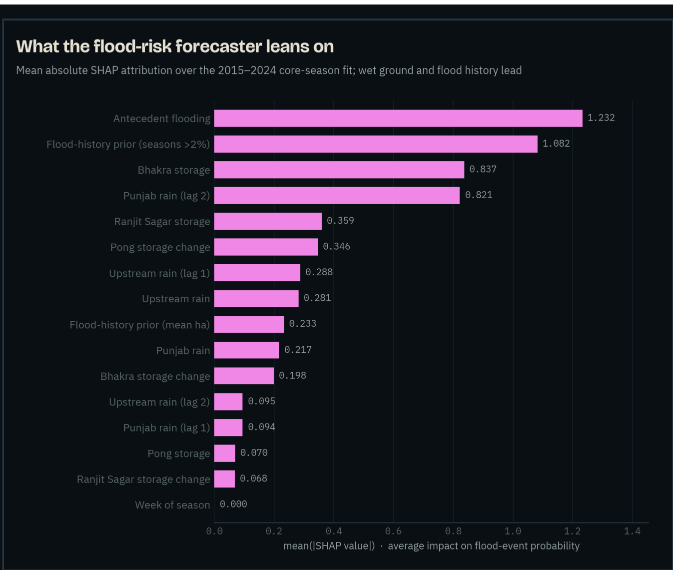
> The headline result. Trained only on 2015–2024 and shown 2025's predictors, the model flagged all five districts our SAR atlas ranked most-flooded, a ranking the government's ground girdawari later corroborated (ρ = 0.72 all districts / 0.56 named), with Kapurthala #1 statewide (P = 0.72 in the pre-declared flood windows; two districts clear P ≥ 0.5; Amritsar is a rank-5 flag at low probability, stated rather than rounded up). The forecasting claim is the pre-crest lead: four of five flagged by Aug 14–24, ~10 days before the Aug 26–27 dam-release peak. We state the critique before a reviewer can: SHAP's top drivers are antecedent flooding and flood history, persistence-aware by design. The pre-crest lead, not post-event probabilities, is the forecast skill.

The ablation we ran on ourselves (pre-declared; both expectations failed; shipped verbatim). *Does the dam signal carry skill?* SHAP ranks Bhakra storage #3 in attribution. But delete all six reservoir features and the 2025 hindcast is unchanged (5/5, Kapurthala #1, same ~10-day lead) while pooled precision even improves. The dam signature is attribution rather than load-bearing skill, and the dam story lives in the headroom analysis, where the evidence supports it. Operationally that is a feature: the BBMB dams stopped reporting centrally in July 2025, so the live 2026 nowcast must run reservoir-blind, and the ablation proves that configuration loses nothing. *Does it beat naive persistence?* On pooled average precision, no (0.31 vs 0.27, published). The model wins where forecasting lives: ROC-AUC 0.946 vs 0.936 and the pre-crest window, 4/5 vs 3/5 early flags. Meteorology-only collapses (3/5, PR-AUC 0.11). Antecedent flooding is the load-bearing signal, which is why the 6-hourly monitor and the forecaster are one system.

Vs Google Flood Hub: river-stage forecasts on ungauged-basin technology; under-models dam-*regulated* rivers, no district crop risk. Sailaab is impact-native and regulation-aware in its analytics; by ablation its detection does not depend on the now-dark reservoir feed. Complementary, not competing.



Why it flooded: ~10× rain spike (26 Aug) over near-full reservoirs → forced releases (Ranjit Sagar 1.73 lakh cusecs, 27 Aug) → inundation. IMD grids + CWC/BBMB records, including the reporting gap.



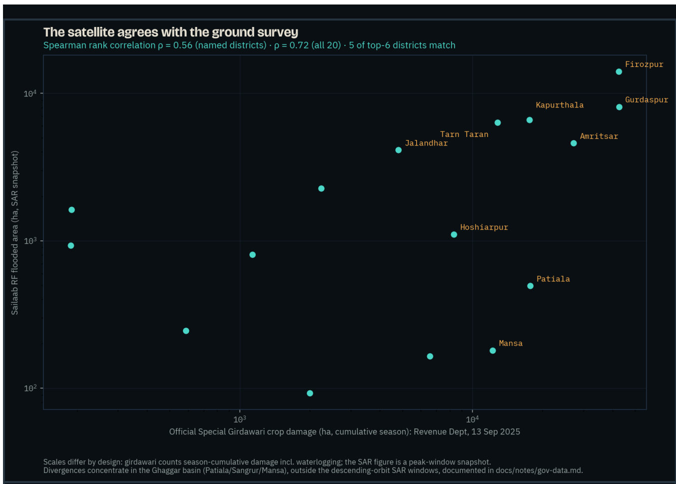
Why the model says what it says: SHAP attribution: wet ground and flood history dominate; Bhakra storage ranks #3 (ablation-tested above: attribution, not load-bearing skill).

04 Validated three independent ways, failures kept

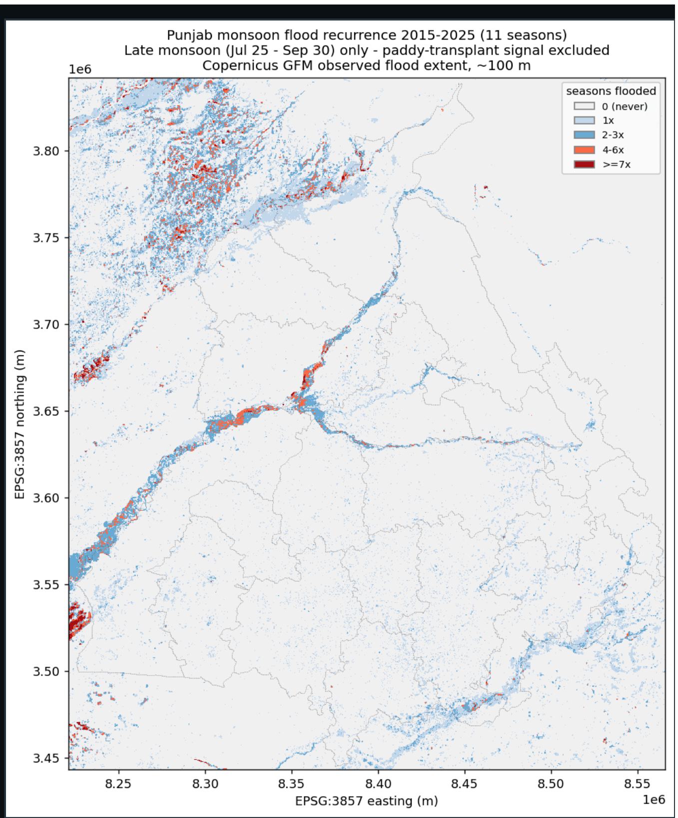
Every satellite step is gated by a pre-declared checkpoint: the expected numeric band is written into the verification log *before* the run, then the actual, then PASS/FAIL. The log ships in the repo with its failures intact (e.g., the NRSC Sangrur-2023 anchor fails on its exact date because no satellite pass existed that day; the same event registers at 33,771 ha in adjacent windows). Validation is triangulated:

- Copernicus GFM (quantitative): fresh-point confusion matrices; RF sits inside its Tier-A/GFM parent envelope on both precision and recall.
- Sentinel-2 optical truth points (independent physics that breaks SAR-chain circularity): 237 photo-interpreted NDWI points confirm 81.7% of RF flood pixels as standing water two weeks post-peak (pre-declared gate ≥60%); all pre-declared precision/recall orderings confirmed.
- ISRO NDEM sheets (visual): approximate-extent cross-check, stated as such; no pixel agreement claimed.

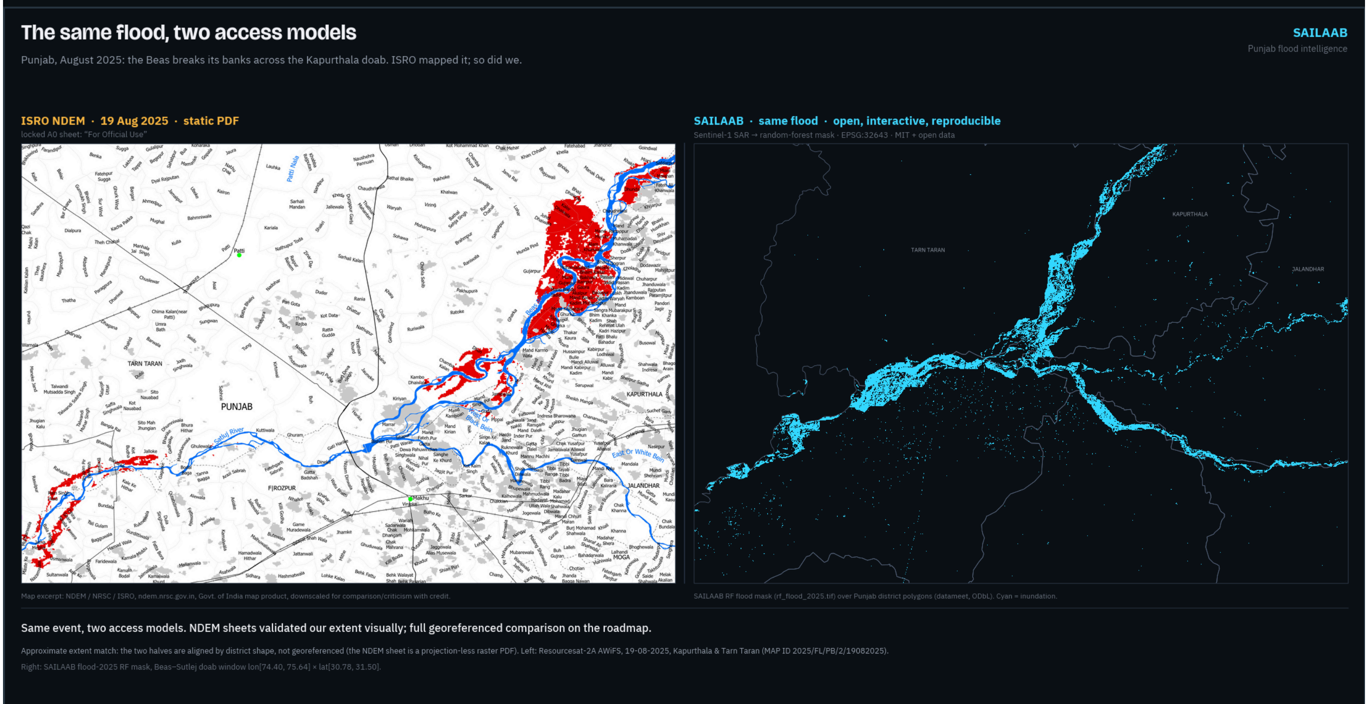
> The ground survey agrees. Against the Revenue Department's Special Girdawari district table (13 Sep 2025), our satellite ranking scores Spearman ρ = 0.72 across all 20 districts (0.56 over the 16 named), with 5 of the top-6 districts identical; divergences concentrate in the Ghaggar basin, outside our SAR windows, and are documented. ₹ value-at-risk was rebuilt on the government's own DES district paddy yields (₹523 crore, within 4% of the independent estimate); the per-pixel submergence-duration atlas (51,202 ha under water for 7 days or more) refines damage to a duration-weighted ₹351 crore. Independent context: IMD's own grids put Aug 20–Sep 5 2025 rainfall at +9.7σ (rank 1 of 11 years), and a pre-registered 65-year Mann-Kendall analysis shows the loading largely stationary except a significantly rising frequency of extreme-wet days upstream (p=0.017); 2025 was a compound timing event, not merely a rainfall record. A pre-declared headroom analysis quantifies the management component honestly: on the eve of the releases, Pong and Ranjit Sagar held +1.0 and +0.45 BCM more than their own decade-median filling practice (Bhakra was near-median), about 1.68 BCM less pre-positioned buffer, worth one to four days of peak release. Real, but second-order to the rain.



Satellite vs ground survey: district ranks vs the Special Girdawari table.



Punjab's first public flood-frequency atlas (2015–2025, transplant-calibrated).



Same flood, two access models: ISRO NDEM's locked PDF sheet (left) vs Sailaab's open, interactive, reproducible map (right). NDEM sheets validated our extent visually; full georeferencing is on the roadmap. Map excerpt: NDEM/NRSC/ISRO.

05 Running right now

The live monitor is not a plan. It executes on GitHub's public runners every 6 hours with zero secrets: it detects each new Sentinel-1 pass over Punjab, computes district flood areas, renders trilingual alerts, and commits the state publicly. The public app is interactive: an every-district map with five switchable layers (live observed, live model risk, 2025 flood, decade frequency, hindcast risk), click-through district panels (flooded ha, crop loss, ₹ at risk, recurrence, hindcast P, official girdawari figure), a before/after satellite swipe, a "replay 2025" scrubber that shows model risk rising before the water arrives, and a full ਪੰਜਾਬੀ/हिन्दी interface toggle. From 25 July 2026, inside judging week, the forecaster also publishes live current-window district probabilities on every CI cycle (it activates only in its trained post-transplant domain; the countdown is shown honestly until then).

06 Impact, users, SDGs

- SDMA / district EOCs: recurrence zones + live district km^2 for pre-positioning and evacuation priority.
- Revenue Dept / girdawari: flood-extent ∩ cropland per tehsil turns weeks of manual survey into verification; 20 print-ready district flood briefs ship in the repo.
- Insurers (PMFBY), banks, farmers: independent reproducible loss extents; vernacular alerts; every map open.
- Worked example (one click in the app): Sultanpur Lodhi tehsil, Kapurthala. Its district was flagged #1 by the model in the Aug 14–24 window; 4,733 ha flooded (4,028 ha cropland); it has flooded in 5 of the last 11 monsoons. Its girdawari list and a ਪੰਜਾਬੀ alert are one click away: the relief workflow, demonstrated on real 2025 data. Outreach to PSDMA and the worst-hit District Collectors is drafted for dispatch with this submission.
- SDGs: 2 (crop-loss quantification), 11 (resilient communities), 13 (climate adaptation).

07 Originality

(1) First open ML-ready flood mapping of Indian-Punjab 2025 (the published RF precedent covered the Pakistani side); (2) Punjab's first public flood-frequency atlas and first submergence-duration atlas; (3) a self-labeling SAR→ML loop that manufactures its own decade of training data, with the rice-transplant contamination discovery published; (4) a district forecaster with a demonstrated ~10-day pre-crest lead, girdawari-corroborated and ablation-stress-tested, both failed expectations published; (5) an end-to-end account-free architecture, so every number re-runs without an account; (6) the quantified forecast gap: Punjab is absent from CWC's own station network, and Sailaab is the district-level layer that gap leaves open. The negative results (a challenger model that lost; conformal intervals failing on the record event; a U-Net losing to the target-domain RF; the ablation's two failed expectations) are published with the same prominence as the wins.

08 Roadmap & openness

Multi-state scale-out (the pipeline is a bounding box + district file), village-circle girdawari integration, Bhashini voice alerts, GPU-scale models via the IndiaAI Compute programme, EOS-04 Indian-SAR cross-validation via NRSC Bhoonidhi, and official partnerships. Code MIT; maps and tables CC-BY-4.0; 459 automated tests; the method paper, data-source registry (every dataset: URL, license, access date) and the full verification log live in the repository.

11 monsoons · 467 flood days · 105,183 ha mapped · 91 tehsils scored · ρ=0.72 vs the government girdawari · 3 languages · 0 CWC forecast stations in Punjab · 0 logins required · github.com/bakathefish/Flood · bakathefish.github.io/Flood ·